| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 2(d) | 10.8 or 10.9 or 10.84 to 10.85... | **4** | **M3** for [sin =] $\frac{their\,(c)}{\sqrt{15^2+5.7^2}}$<br>or $\left[\tan =\right]\frac{their\,(c)}{\sqrt{\left(5.7\times cos32\right)^2+15^2}}$<br><br>or **M2** for $15^2+5.7^2$ or $\left(5.7\times\cos 32\right)^2+15^2$ oe<br><br>or **M1** for recognition of correct angle |
| 2(e) | 136 or 136.0... | **3** | **M2** for $938\times145\times\frac{1000}{1000000}$ oe<br>or **M1** for figs 136 or 13601 |
| 3(a)(i) | 55.87 | **4** | **M1** for midpoints soi<br><br>**M1** for use of $\sum fm$ where $m$ is in the correct interval including boundaries<br><br>**M1** (dep on 2nd M1) for $\sum fm \div 1000$ |
| 3(a)(ii) | $\frac{177}{500}$ cao | **2** | **M1** for $\frac{154+200}{1000}$ oe |
| 3(b)(i) | 25000 | **1** | |
| 3(b)(ii) | $2.473\times10^4$ | **1** | |
| 3(c)(i) | 166 650 or 165816 nfww | **3** | **M2** for (500 + 5) × '320 to 340'<br>or '500 to 510' × (320 + 10)<br><br>or **M1** for 500 − 5 or 500 + 5 or 320 $-10$ or 320 $+10$<br><br>Alternative method<br>**M2** for 504 × '320 to 340'<br>or '500 to 510' × 329<br><br>or **M1** for 504 or 329 |
| 3(c)(ii) | 285 or 286 nfww | **2** | **M1** for 800 $-10$ |
| 4(a)(i) | 96 | **2** | **M1** for $\frac{1}{2}\times24\times8$ |
| 4(a)(ii) | 18.4 or 18.43... | **2** | **M1** for $\tan\left[x\right]=\frac{8}{24}$ oe |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 4(b) | 622 or 622.0 to 622.1.... | **2** | **M1** for $[\frac{1}{2}\times]\ \pi\times 6^2\times 11$ or $\frac{1}{2}\times\pi\times 6^2[\times 11]$ |
| 4(c)(i) | 246 or 246.2 to 246.3... | **5** | **M4** for $15\times 20-4\times 4-\frac{270}{360}\times\pi\times 4^2$ oe<br>OR<br>**M2** for $\frac{270}{360}\times\pi\times 4^2$ oe<br>or **M1** for $k\times\pi\times 4^2$, where $k\le 1$<br><br>**M1** for $15\times 20$ or $4\times 4$ oe |
| 4(c)(ii) | 80.8 or 80.9 or 80.84 to 80.85... | **3** | **M1** for $15+20+11+16$ oe<br>**M1** for $\frac{3}{4}\times 2\times\pi\times 4$ oe |
| 5(a)(i)(a) | 25 | **1** | |
| 5(a)(i)(b) | 17 to 18 | **1** | |
| 5(a)(i)(c) | 12 | **2** | **B1** for 148 seen |
| 5(a)(i)(d) | 30 | **2** | **B1** for 104 seen |
| 5(a)(ii)(a) | correct diagram or correct for *their* median and LQ | **3** | **B1** for whiskers at 1 **and** at 70<br><br>**B1** for with median **and** LQ at *their* **(a)(i)(a)** and **(a)(i)(b)**<br><br>**B1** for UQ at 34<br>Maximum 2 marks if diagram incorrect<br>If 0 scored **SC1** for *their* 5 correct ages plotted |
| 5(a)(ii)(b) | 50 | **1** | |
| 5(b) | correct histogram | **3** | **B1** for each correct block<br>width 10 height 3.7<br>width 20 height 1.2<br>width 30 height 2<br><br>If 0 scored **SC1** for correct frequency densities 3.7, 1.2, 2 oe |